

Supporting Information

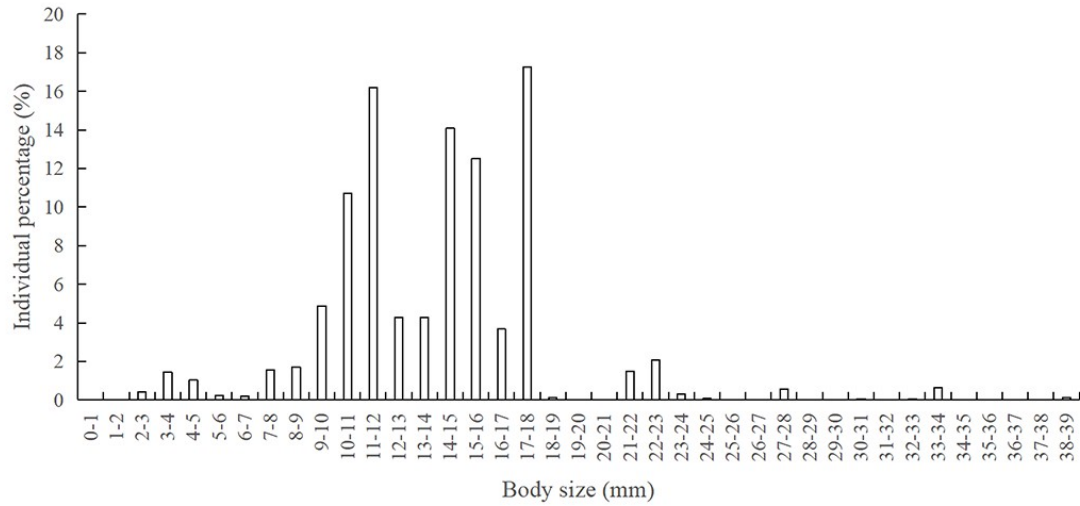


Fig. S1. Distribution histogram of carabid body size.

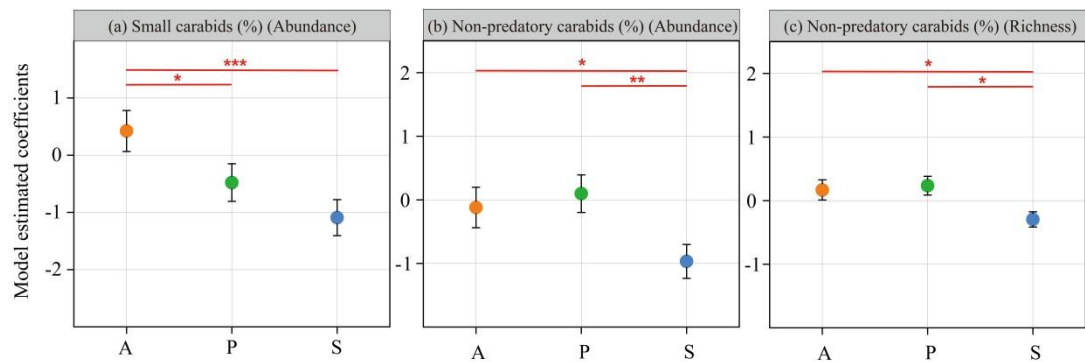


Fig. S2. Estimated coefficients from generalized linear mixed models showing the differences between assemblages in different land use types (A: annual crops; P: perennial crops; S: semi-natural habitats): (a) proportion of small carabids abundance (b) proportion of non-predatory carabids abundance (c) proportion of non-predatory carabids species richness; error bars refer to the SE of estimates, and asterisks indicate a significant difference based on the models (* ≤ 0.05 ; ** ≤ 0.01 ; *** ≤ 0.001).

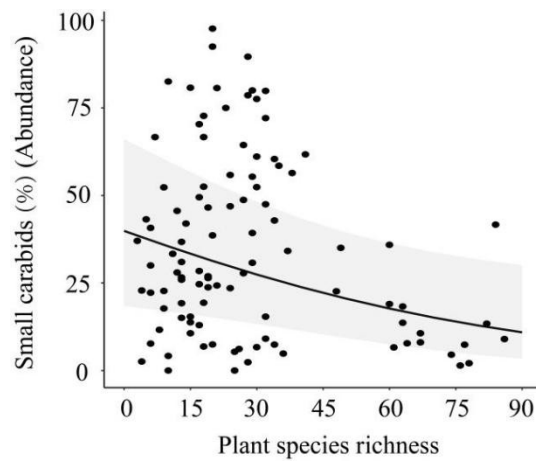


Fig. S3. Effects of the plant species richness on the proportion of small carabid abundance.

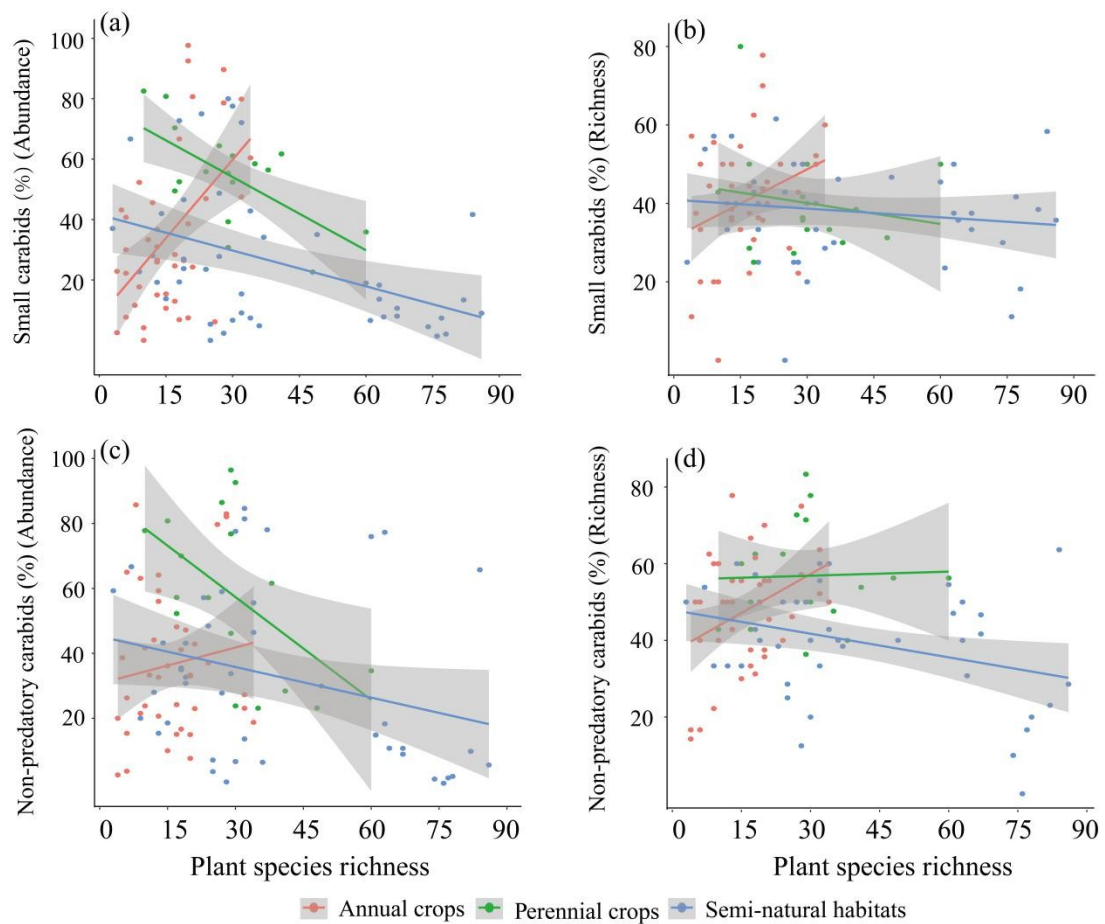


Fig. S4. Effects of land use types and plant species richness on the proportion of small-sized carabids with regards to abundance (a) and species richness (b), and on

the proportion of non-predatory carabids based on abundance (c) and species richness (d).

Table S1. Land use types and sampling time in different areas, and brackets refer to the number of plots and sampling days of every month, respectively.

Study area	Land use type and number of plots			Sampling period	
	Annual crop (plots)	Perennial crop (plots)	Semi-natural land (plots)	Month /days. Month-1	Year
Baiqi	Potato field (2) Maize field (1) Oat field (1)	/	Plantation forest (4)	May~October /6	2006~2007
Fangshan	/	Walnut planting (4)	Grassland (4) Plantation forest (4) Alpine meadow (4)	May~October /6	2011~2012
Gaojiaying	Maize field (4)	Orchard (4)	Grassland (4)	May~October /6	2006~2007
Haidian	Maize field (4)	Orchard(4)	Woodland (4)	May~October /6	2006
Miyun	Wheat field (4) Peanut field (4)	Orchard (4)	Field margin (4) Windbreak (4) Woodland (4)	May~October/ 6	2010
Qianjiang	Rainfed field (4) Paddy field (8) Vegetable field (8)	/	Dyke grassland (5) Field margin (5) Windbreak (6) Woodland (4)	May-October /4	2013
Quzhou	Wheat/Maize field (4) Cotton field (4)	Orchard (4)	Woodland (3) Grassland (3)	May-October /10	2005~2006
Shizigou	Cabbage field (2) Potato field (2)	/	Plantation forest (4) Grassland (4)	May~October /6	2006~2007
Yanqing	Maize field (4)	Orchard (4)	Woodland(4)	May-October /6	2006~2007
Total number of plots	56	24	74	/	/

Table S2. Plant species richness in different land use types.

Habitat type	Community structure	Dominant species	Mean±SE	Min	Max
Annual cops	Herb layer	<i>Zea mays</i> , <i>Arachis hypogaea</i> , <i>Digitaria sanguinalis</i> , <i>Glycine max</i> , <i>Brassica pekinensis</i> , <i>Echinochloa crusgalli</i> , <i>Acalypha australis</i> , <i>Triticum aestivum</i> , <i>Gossypium hirsutum</i> , <i>Solanum tuberosum</i>	14.73±1.26	3	43
Perennial crops	Herb layer	<i>Amaranthus retroflexus</i> , <i>Humulus scandens</i> , <i>Digitaria sanguinalis</i> , <i>Pennisetum alopecuroides</i> , <i>Artemisia scoparia</i> , <i>Kochia scoparia</i> , <i>Chenopodium glaucum</i> , <i>Salsola collina</i> , <i>Artemisia argyi</i> , <i>Hibiscus trionum</i>	22.54±1.93	7	56
	Tree layer	<i>Armeniaca vulgaris</i> , <i>Juglans regia</i> , <i>Malus asiatica</i> , <i>Amygdalus persica</i> , <i>Malus domestica</i> , <i>Prunus salicina</i> , <i>Pyrus sorotina</i>	2.42±0.26	4	1
Semi-natural habitats	Herb layer	<i>Artemisia capillaris</i> , <i>Carex humilis</i> , <i>Cynodon dactylon</i> , <i>Chenopodium glaucum</i> , <i>Humulus scandens</i> , <i>Carex duriuscula</i> , <i>Digitaria sanguinalis</i> , <i>Artemisia annua</i> , <i>Eremochloa ophiuroides</i> , <i>Cyperus rotundus</i>	32.77±2.87	3	87
	Shrub layer	<i>Vitex negundo</i> , <i>Wikstroemia chamaedaphne</i> , <i>Spiraea trilobata</i> , <i>Cayratia japonica</i> , <i>Lespedeza inschanica</i> , <i>Clerodendrum bungei</i>	2.06±0.31	0	13
	Tree layer	<i>Larix principis-rupprechtii</i> , <i>Populus × beijingensis</i> , <i>Populus cathayana</i> , <i>Robinia pseudoacacia</i> , <i>Ulmus pumila</i> , <i>Populus tomentosa</i> , <i>Ulmus davidiana</i> , <i>Juglans mandshurica</i> , <i>Metasequoia glyptostroboides</i>	2.48±0.23	0	8

Table S3. Parameters of plant species richness in different land use types.

Land use types		Mean±SE
Annual crops	Cabbage field	33.00±10.00
	Potato field	26.80±2.87
	Vegetable field	20.00±1.15
	Maize field	17.31±1.71
	Cotton field	14.75±1.31
	Wheat field	11.50±2.15
	Rainfed field	9.60±1.81
	Paddy field (wheat field and soybean field)	5.33±0.67
	Peanut field	5.25±1.31
Perennial crops	Walnut planting	28.75±0.63
	Orchard	26.55±2.90
Semi-natural habitats	Grassland	62.27±6.87
	Plantation forest	41.25±6.04
	Woodland	30.36±1.79
	Alpine meadow	26.75±3.09
	Windbreak	22.71±3.08
	Field margin	15.29±1.39
	Dyke grassland	6.00±1.73

Table S4. Species list of carabid beetles with number of individuals and trophic levels
(Small < 12 mm and large ≥ 12 mm).

	Specie	Body size (mm)	Annual crops	Perennial crops	Semi-natural habitats	trophic level
small	<i>Acupalpus dubius</i> Schilsky, 1888	3.62	1	0	1	Predatory
	<i>Agonum</i> sp1.	8.64	1	0	0	/
	<i>Agonum</i> sp2.	10.02	7	1	40	Predatory
	<i>Amara alacris</i> Tschitscherine, 1899	5.75	0	1	0	Non-predatory
	<i>Amara aurichalcea</i> Germar, 1824	8.75	8	0	0	Non-predatory
	<i>Amara chalcites</i> Dejean, 1828	9.31	12	0	2	Non-predatory
	<i>Amara communis</i> Panzer, 1796	7.33	10	0	15	Non-predatory
	<i>Amara harpaloides</i> Dejean, 1828	11.41	1	0	6	Non-predatory
	<i>Amara majuscula</i> Chaudoir, 1850	7.20	1	0	4	Non-predatory
	<i>Amara</i> sp1.	8.32	2	0	0	Non-predatory
	<i>Amara</i> sp2.	7.35	3	0	12	Non-predatory
	<i>Amara</i> sp3.	9.14	4	0	2	Non-predatory
	<i>Amara tumida</i> Morawitz, 1862	6.53	0	0	3	Non-predatory
	<i>Anisodactylus punctatipennis</i> Morawitz, 1862	11.30	19	0	5	Non-predatory
	<i>Asaphidion semilucidum</i> Motschulsky, 1862	4.69	12	19	7	Non-predatory
	<i>Bembidion difforme</i> Motschulsky, 1844	4.00	4	0	1	Predatory
	<i>Bembidion incidiosum</i>	4.91	48	1	0	Predatory
	<i>Bembidion pogonoides</i> Bate, 1883	6.54	3	0	0	Predatory
	<i>Bradycellus</i> sp.	6.06	2	0	0	Non-predatory
	<i>Callistomimus modestus</i> Schaum, 1863	4.96	1	0	0	Predatory
	<i>Chlaenius inops</i> Chaudoir, 1856	11.01	3	0	0	Predatory
	<i>Chlaenius pericallus</i> Redtenbacher, 1867	11.68	2	0	0	Predatory
	<i>Cymindis daimio</i> Bates, 1873	9.00	0	0	2	Predatory
	<i>Cymindis</i> sp.	9.54	1	0	1	Predatory
	<i>Dyschirius</i> sp.	3.30	0	0	2	Predatory
	<i>Harpalus aequicollis</i> Motschulsky, 1844	11.55	3	0	2	Non-predatory
	<i>Harpalus bungii</i> Chaudoir, 1844	8.57	18	37	29	Non-predatory
	<i>Harpalus davidi</i> Tschitscherine, 1897	10.80	0	1	1	Non-predatory
	<i>Harpalus davidianus</i> Tschitscherine, 1903	9.85	1	0	14	Non-predatory
	<i>Harpalus griseus</i> Panzer, 1796	11.33	68	169	30	Non-predatory
	<i>Harpalus melaneus</i> Bates, 1878	11.97	1	0	2	Non-predatory
	<i>Harpalus pallidipennis</i> Morawitz, 1862	10.12	521	108	289	Predatory
	<i>Harpalus pastor</i> Motschulsky, 1844	11.88	317	266	244	Non-predatory
	<i>Harpalus</i> sp.	7.39	4	0	49	Non-predatory
	<i>Harpalus suenisoni</i> Kataev, 1997	11.50	3	0	2	Non-predatory
	<i>Harpalus tinctulus</i> Bates, 1873	7.40	0	1	0	Non-predatory
	<i>Harpalus tridens</i> Morawitz, 1862	11.85	2	0	326	Non-predatory
	<i>Lebia</i> sp.	4.64	0	0	1	Predatory

	<i>Lesticus magnus</i> Motschulsky, 1860	7.79	17	2	17	Predatory
	<i>Loricera pilicornis</i> Fabricius, 1775	7.79	0	0	2	Predatory
	<i>Loxoncus circumcinctus</i> Motschulsky, 1857	9.16	4	0	0	Predatory
	<i>Licinus setosus</i> Sahlberg, 1880	8.00	4	0	5	/
	<i>Mastax poecila</i> Schaum, 1863	3.90	1	0	1	Non-predatory
	<i>Oxycentrus jelineki</i> Ito, 2006	8.32	15	1	3	Predatory
	<i>Oxycentrus</i> sp.	6.94	5	0	5	Predatory
	<i>Panagaeus davidi</i> Fairmaire, 1887	10.8	1	0	0	Predatory
	<i>Peronomerus auripilis</i> Bates, 1883	9.50	0	1	0	Predatory
	<i>Pheropsophus occipitalis</i> Macleay, 1825	8.85	3	1	0	Predatory
	<i>Pterostichus haptoderoides</i> Tschitschérine, 1889	7.46	2	0	2	Predatory
	<i>Pterostichus kiangsu</i> Jedlicka, 1965	11.82	0	0	4	Predatory
	<i>Pterostichus microcephalus</i> Motschulsky, 1860	9.50	111	217	14	Predatory
	<i>Pterostichus sulcitaris</i> Morawitz, 1862	8.93	5	0	11	Predatory
	<i>Stenaspis castaneipennis</i> Dupont, 1838	10.00	0	0	9	/
	<i>Stenolophus difficilis</i> Hope, 1845	5.38	17	0	4	Predatory
	<i>Syntomus pallipes</i> Dejean 1825	3.30	25	2	1	Predatory
	<i>Synuchus major</i> Lindroth, 1956	8.21	0	0	1	Predatory
	<i>Synuchus nordmani</i> Morawitz, 1862	9.25	32	3	24	Predatory
	<i>Synuchus</i> sp.	8.21	3	1	8	Predatory
	<i>Tachus gradatus</i>	3.04	83	14	2	Non-predatory
	<i>Tachys</i> sp1.	2.62	32	0	3	Non-predatory
	<i>Tachys</i> sp2.	2.10	2	0	0	Non-predatory
large	<i>Anisodactylus signatus</i> Panzer, 1797	12.25	7	6	3	Non-predatory
	<i>Amara gigantea</i> Motschulsky, 1844	18.31	0	2	7	Non-predatory
	<i>Amara macronota</i> Solsky, 1875	13.23	7	23	15	Non-predatory
	<i>Brachinus scotomedes</i> Redtenbacher, 1867	14.00	7	0	3	Predatory
	<i>Calosoma denticolle</i> Gebler, 1833	22.50	0	0	1	Predatory
	<i>Calosoma lugens</i> Chaudoir, 1869	23.00	1	3	0	Predatory
	<i>Carabus brandti</i> Faldermann, 1835	27.28	16	15	11	Predatory
	<i>Carabus canaliculatus</i> Adams, 1812	24.67	0	0	7	Predatory
	<i>Carabus conciliator</i> Fischer von Waldheim, 1822	30.26	1	0	0	Predatory
	<i>Carabus crassesculptus</i> Kraatz, 1881	23.03	3	0	20	Predatory
	<i>Carabus kruberi</i> Fischer von Waldheim, 1822	21.00	6	0	76	Predatory
	<i>Carabus latreillei</i> Fischer von Waldheim, 1820	22.30	0	3	10	Predatory
	<i>Carabus manifestus</i> Kraatz, 1881	21.03	0	0	31	Predatory
	<i>Carabus smaragdinus</i> Fischer von Waldheim, 1823	33.63	26	14	19	Predatory
	<i>Carabus vladimirskyi</i> Dejean, 1830	22.87	0	1	142	Predatory
	<i>Carabus elysii</i> Thomson, 1856	39.00	3	0	10	Predatory
	<i>Chlaenius aspericollis</i> Bates, 1873	13.76	36	0	10	Predatory
	<i>Chlaenius bengalensis</i> Chaudoir, 1856	15.86	4	0	4	Predatory
	<i>Chlaenius bimaculatus</i> Dejean 1826	13.60	2	0	7	Predatory

<i>Chlaenius costiger</i> Chaudoir, 1856	22.23	1	0	5	Predatory
<i>Chlaenius leueops</i>	13.24	1	0	23	Predatory
<i>Chlaenius micans</i> Fabricius, 1792	15.96	513	11	64	Predatory
<i>Chlaenius pallipes</i> Gebler, 1823	14.17	31	2	2	Predatory
<i>Chlaenius posticalis</i> Motschulsky, 1854	14.50	229	47	181	Predatory
<i>Chlaenius touzalini</i> Andrewes, 1920	14.75	6	0	0	Predatory
<i>Chlaenius virgulifer</i> Chaudoir, 1876	13.50	2	2	1	Predatory
<i>Curtonotus giganteus</i> Motschulsky, 1844	21.30	1	10	13	Non-predatory
<i>Dolichus halensis</i> Schaller, 1783	17.59	125	283	223	Predatory
<i>Harpalus brevicornis</i> Germar, 1824	15.47	6	2	1	Non-predatory
<i>Harpalus crates</i> Bates, 1883	12.57	3	3	0	Non-predatory
<i>Harpalus chalcatus</i> Bates, 1873	12.74	72	24	71	Non-predatory
<i>Harpalus coreanus</i> Tschitscherine, 1895	14.35	0	1	0	Non-predatory
<i>Harpalus corporosus</i> Motschulsky, 1861	15.03	76	69	11	Non-predatory
<i>Harpalus eous</i> Tschitscherine, 1901	15.29	2	0	0	Non-predatory
<i>Harpalus lumbaris</i> Mannerheim, 1825	13.04	34	1	5	Non-predatory
<i>Harpalus roninus</i> Bates, 1873	17.60	8	0	0	Predatory
<i>Harpalus rubripes</i> Duftschmid, 1812	13.35	0	0	1	Non-predatory
<i>Harpalus simplicidens</i> Schauburger, 1929	12.67	119	66	16	Non-predatory
<i>Harpalus sinicus</i> Hope, 1845	13.73	54	6	147	Non-predatory
<i>Harpalus ussuriensis</i> Chaudoir, 1863	13.73	5	0	3	Non-predatory
<i>Harpalus vicaius</i> Harold, 1878	17.60	5	1	0	Predatory
<i>Nebria livida angulata</i> Bänninger, 1949	16.65	2	0	0	Predatory
<i>Pheropsophus jessoensis</i> Morawitz, 1862	17.37	300	0	353	Predatory
<i>Planetes puncticeps</i> Andrewes, 1919	13.63	0	0	1	Predatory
<i>Pristosia</i> sp.	13.76	1	0	4	Predatory
<i>Pseudotaphoxenus mongolicus</i> Jedlička, 1953	15.00	7	2	306	Predatory
<i>Pseudotaphoxenus rugipennis</i> Faldermann, 1835	22.07	3	0	16	Predatory
<i>Pterostichus acutidens</i> Fairmaires, 1888	17.46	15	0	24	Predatory
<i>Pterostichus adstrictus</i> Eschscholtz, 1823	15.79	0	0	63	Predatory
<i>Pterostichus fortipes</i> Chaudoir, 1850	14.44	40	94	640	Predatory
<i>Pterostichus gebleri</i> Dejean, 1828	16.44	64	52	217	Non-predatory
<i>Pterostichus interruptus</i> Dejean, 1828	18.02	0	0	1	Predatory
<i>Scarites acutidens</i> Chaudoir, 1855	32.09	4	0	2	Non-predatory
<i>Scarites sulcatus</i> Olivier, 1795	27.62	7	0	2	Predatory
<i>Scarites terricola</i> Bonelli, 1813	17.61	127	18	92	Non-predatory
<i>Tachys</i> sp3.	18.31	1	0	0	/
<i>Tachys</i> sp4.	22.50	1	0	0	/
<i>Trigonognatha jeachi</i> Sciaky 1995	22.22	0	2	5	Non-predatory